PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: AdPilot

Version: 1.0

1. Project Overview

- **Problem Statement:**

  E-commerce sellers in Southeast Asia especially small and medium-sized businesses operating on platforms such as Shopee, Lazada, and TikTok Shop usually face a critical gap between having products to sell and knowing how to advertise them effectively. According to the e-Conomy SEA 2023 report by Google, Temasek, and Bain & Company, the SEA e-commerce market reached USD 159 billion in GMV in 2023 and is projected to grow to USD 295 billion by 2028. Despite this enormous opportunity, the majority of SME sellers are unable to capitalise on it fully due to a lack of structured advertising strategy.

  Most sellers rely on guesswork when making advertising decisions choosing platforms, formats, and posting times without data to support them. A 2022 study by Lazada's internal seller research division found that over 65% of SME sellers on their platform made advertising decisions based on intuition or peer advice rather than data. This results in wasted advertising budget, unsold inventory, and missed seasonal sales opportunities. Meta Business (2023) estimated that SMEs in SEA waste approximately 37–42% of their digital ad spend due to poor targeting and timing decisions.

  The problem is compounded by three structural barriers:

  - First, sellers have no tool that connects to their inventory situation. What they have in stock, what is overstocked, what is new to advertising urgency.
  - Second, SEA e-commerce is highly seasonal (11.11, Hari Raya, Chinese New Year, Valentine's Day), and most sellers lack a system to proactively align their advertising with upcoming peak events.

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- Third, hiring a marketing professional in Malaysia costs between RM 3,000–8,000 per month (MDEC, 2023), making professional ad strategy inaccessible for most SMEs.

The result is a structural competitive disadvantage: SME sellers compete on the same platforms as enterprise brands that have full marketing teams, agency partners, and data tools, without any equivalent resource.

- **Target Domain:**

  Enabling data-driven advertising decisions for small and medium size's e-commerce sellers in Southeast Asia by leveraging Z. AI's GLM as the core intelligence engine.

- **Proposed Solution Summary:**

  AdPilot is an AI-powered advertising decision and content generation platform built entirely on Z.ai's GLM models. It analyzes a seller's inventory, cross-references real-time market trends and seasonal event calendars, and generates a ranked list of advertising strategy recommendations.

  Each recommendation outlines:

Which platform to advertise on (e.g. Tiktok, Facebook)
What ad format to use (e.g. Picture/Video/Text)
What pricing strategy to apply
Who to target
When to post

  The seller's role shifts from spending hours researching, planning, and writing ads to simply reviewing the AI's reasoning and selecting the strategy they want to execute.

2. Background & Business Objective

- **Background of the Problem:**

For many years, e-commerce advertising followed a manual, expertise-heavy process. A seller had to independently research which products to promote, study platform algorithms, monitor seasonal trends, assess competitor pricing, and write ad copy. All without a structured system to guide them. This required significant time investment and marketing domain knowledge that most SME sellers in market simply do not have. Platforms like Shopee Ads and TikTok for Business provide the tools to execute advertising campaigns, but they do not tell sellers what to advertise, where, when, or to whom. They assume the seller already has a strategy and that is the gap AdPilot fills.

The Journal of Small Business Management (2022) found that SME and e-commerence owners spend an average of 6.3 hours per week on marketing-related tasks, with ad planning and content creation accounting for the majority. This challenge is even greater for first-time entrepreneurs who lack formal business or marketing education. The high time commitment, cost of hiring marketing experts, and complexity of managing ads across multiple platforms often cause e-commerce sellers to either overlook advertising opportunities or spend inefficiently on campaigns that deliver poor results.

- **Importance of Solving This Issue:**
  AdPilot reduces advertising strategy development from hours of manual research to under 60 seconds, allowing sellers to act on opportunities faster than ever before—without needing a dedicated marketing team. By removing the expertise barrier, AdPilot enables any seller to create smarter advertising strategies and compete more effectively with larger brands on the same platforms. The SEA e-commerce market's projected growth to USD 295 billion by 2028 (Google/Temasek/Bain, 2023) means the sellers who will capture this growth are not necessarily those with the best products, but those who market them most effectively. AdPilot directly addresses this by giving every SME seller access to enterprise-grade marketing intelligence at an accessible price point.

- **Strategic Fit / Impact:**

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AdPilot system is built on Z.AI's GLM as its sole intelligence engine. GLM interprets structured inventory data, performs multi-factor contextual reasoning across trends and seasonal calendars, generates ranked strategy recommendations with transparent explanations, and produces platform-tailored ad content within a single agentic pipeline. Without GLM, AdPilot would not be able to deliver its core functionality.

By reducing the time, cost, and complexity of advertising, AdPilot helps Southeast Asia's SME sellers make smarter marketing decisions, compete more effectively with larger brands, and capture growth in the region's rapidly expanding digital economy.

3. Product Purpose

3.1. Main Goal of the System:

To provide an AI-powered advertising partner for SME e-commerce sellers in Southeast Asia by transforming raw inventory data into actionable advertising strategies and ready-to-publish content, powered entirely by Z.ai's GLM. AdPilot enhances it by giving small sellers access to the same level of marketing intelligence and strategic support that larger brands typically have.

3.2. Intended Users (Target Audience):

- SME E-Commerce Sellers
- Social Commerce Creators
- SME-Focused Marketing Agencies
- First-Time Entrepreneurs / New Market Entrants

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4. System Functionalities

4.1. Description:

AdPilot operates as a high-speed AI decision engine for e-commerce advertising. Through its Inventory-to-Strategy-to-Content pipeline, Z.ai's GLM analyzes a seller's product inventory, evaluates key factors such as stock levels, market trends, and Southeast Asia's seasonal demand cycles, and generates a complete advertising strategy with ready-to-publish content. The system is structured as a four-step agentic flow in which GLM performs all analytical and generative work

4.2. Key Functionalities:

- **Inventory Intelligence:**
  The seller uploads a CSV file containing product names, stock quantities, categories, and pricing. GLM reads and interprets this structured data to identify which products need promotion, which are overstocked, which are new with no sales history, and which are seasonally relevant to upcoming e-commerce events in SEA.
- **AI Ad Strategy Recommendations:**
  GLM cross-references the inventory analysis with current market trends and the SEA (Southeast Asia) seasonal calendar to generate 3–5 ranked advertising strategy recommendations. Each recommendation covers five dimensions: the platform to advertise on (Shopee, Lazada, TikTok Shop, Instagram, or Facebook), the ad format (short-form video, image carousel, or static image), the pricing strategy (standard pricing, discount, bundle, or flash sale), the target audience profile (age, interest, and buying behaviour), and the best time to post. Every recommendation is accompanied by a plain-language explanation of the reasoning behind each decision, so sellers without marketing expertise can evaluate it confidently.
- **Seller Review & Decision Dashboard:**
  All recommendations are displayed in a clean dashboard as small recommendation cards outlining key details such as platform, ad format, pricing strategy, target audience, and posting time. Sellers can review the AI's reasoning, compare options, and select the strategy they want to proceed with. To maintain human oversight in the decision-making

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process, AdPilot does not automatically execute recommendations. Sellers remain the final decision-makers.

- **Ad Content Generation**

  Based on the selected recommendation and the seller's product details from the CSV, GLM generates complete, ready-to-publish advertising content: an ad headline and body copy, a platform-optimized caption, a curated hashtag set, and posting guidance. The content is tailored to current trends and formats on the selected platform to maximize relevance and engagement.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

AdPilot uses Z.AI's GLM as its sole intelligence engine. GLM was chosen because it can read and interpret structured CSV data directly without pre-processing, reason across multiple factors simultaneously such as inventory state, market trends, seasonal events and produce both analytical recommendations and creative ad content within the same model. It also generates plain-language explanations that any seller can understand without a marketing background. If GLM is unavailable, AdPilot cannot function at all.

4.3.2. Prompting Strategy:

AdPilot uses multi-step agentic prompting, splitting the task into four sequential GLM calls. A single prompt was tested and rejected. It produced inconsistent outputs because the model was handling too many objectives at once. Breaking it into stages allows GLM to focus on one task at a time: Stage 1 classifies the inventory, Stage 2 generates strategy recommendations, Stage 3 is the seller's decision, and Stage 4 generates the ad content. Few-shot examples are included in Stage 2 and Stage 4 to keep outputs structured and on-tone.

4.3.3. Context & Input Handling:

AdPilot accepts CSV files of up to 500 product rows. Files between 200 and 500 rows are processed in chunks of 100 to 150 rows per GLM call, with outputs merged before Stage

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2 begins. Files exceeding 500 rows are rejected before any API call is made, and the seller is guided to filter or split their upload. Malformed or incomplete CSVs — missing product names, non-numeric quantities, empty rows — are caught by a validator before reaching GLM. The system only calls GLM once the input is clean.

4.3.4. Fallback & Failure Behavior:

All GLM outputs at Stage 1 and Stage 2 are validated against a strict JSON schema before moving forward. If a response fails such as whether off-topic, incomplete, or hallucinated, the system automatically retries up to two times with a more explicit prompt. If retries still fail, the seller sees a plain-language error message and no broken content is shown. For Stage 4, a post-generation check confirms the ad copy matches the product name, price, and category from the CSV. Any mismatch triggers a re-generation of that specific element before the content reaches the seller.

5. User Stories & Use Cases

- **User Stories:**
  - "As a Shopee seller with no marketing background, I want to upload my product inventory and instantly receive a data-driven advertising strategy without spending hours researching platforms or writing ad copy."
  - "As a fashion seller preparing for Hari Raya, I want the system to identify which products are most relevant for the festive season so I can launch targeted campaigns early and maximize seasonal sales."
  - "As a seller with 400 units of slow-moving phone cases, I want AI-driven recommendations on the most effective advertising strategy to clear excess inventory and free up capital efficiently."

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- "As a beauty seller launching a new skincare product with no sales history, I want a launch strategy and ready-to-publish ad content so I can generate visibility without hiring a marketing team."

- "As a TikTok seller, I want short-form video scripts and trending hashtags generated for my product so I can publish content immediately without spending hours creating it manually."

- **Use Cases (Main Interactions):**

  - **Inventory-to-Strategy Flow:** The seller uploads a CSV file containing their product inventory. AdPilot's parser validates the data and passes it to GLM. GLM analyses the inventory alongside injected trend and seasonal context, then generates a ranked set of advertising strategy recommendations. Each recommendation is displayed on the seller dashboard with full plain-language reasoning covering platform, ad format, pricing strategy, target audience, and best posting time.

  - **Recommendation Review & Selection:** The seller reviews all recommendations displayed on the dashboard, compares the suggested strategies, and reads the AI-generated reasoning behind each option. They then select the strategy they want to proceed with. To maintain human oversight in the decision-making process. The seller remains responsible for making the final decision. Once selected, the chosen strategy and relevant product data are passed to the content generation stage.

  - **Ad Content Generation:** Based on the selected strategy and product details, GLM generates the complete ad content package: ad headline and body copy, platform-optimized caption, hashtag set, and posting guidance. The seller receives the content in an editable preview, makes any adjustments they wish, and exports it for use in their platform's ad manager.

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6. Features Included (Scope Definition)

- Personal account creation allows sellers to register, log in, and manage their inventory uploads, recommendation history, and generated advertising content in one personalized dashboard.
- Personalized advertising recommendations based on uploaded CSV inventory data, allowing the system to analyze product names, stock levels, categories, and pricing to generate tailored campaign strategies for each seller
- Z.ai GLM-powered inventory classification, including promotional urgency ranking, seasonal relevance detection, and new product identification
- AI-generated strategy recommendation dashboard displaying at least 2 ranked recommendations per session, each with clear reasoning across five advertising dimensions.
- Seasonal and trend-aware recommendation engine that identifies upcoming e-commerce events relevant to the seller's inventory and based on user target region.
- Platform-specific ad content generation for multiple social platform, including ad copy, captions, hashtags, and posting guidance
- Mandatory seller review and strategy selection before content generation

7. Features Not Included (Scope Control)

- Direct integration with Shopee Ads, TikTok Ads Manager, Instagram, and Facebook
- Real-time inventory synchronization with e-commerce stores
- AI-generated product video creatives
- Campaign performance tracking and ROI analytics
- Multilingual content generation
- Editable content preview before export
- Final campaign publishing requires seller approval and manual execution

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8. Assumptions & Constraints

- LLM Cost Constraint

  Each average user session — including one inventory upload, one recommendation set, and one content generation cycle — is estimated to consume approximately 6,000 tokens for inventories of around 50 products. For larger inventories (200–500 products), token usage may increase to approximately 12,000–15,000 tokens due to data chunking and additional processing requirements.

  To keep inference costs manageable at scale, a CSV upload caching mechanism was implemented. Each user's uploaded inventory and its generated output are stored as session history. If the same user uploads a similar or identical CSV in a future session, the system serves the cached result directly without triggering a new GLM call.

- **Technical Constraints**

  1. Inventory input is limited to CSV file uploads only.

  2. Ad content export is currently text-based and image. Video generation is not supported.

- **Performance Constraints**

  1. Larger inventories (200–500 products) may require 25–45 seconds and higher token consumption (12,000–15,000 tokens) because more product entries need to be processed, analyzed, and matched against market trends and seasonal opportunities before recommendations can be generated.

  2. Recommendation quality depends on the accuracy and completeness of uploaded product data. Generic product names such as “Product A” may produce weaker recommendations.

- **User Input Constraints**

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1. The system requires a completed store inventory of CSV with at least a product name and stock quantity for each item.

2. Sellers must review and approve recommendations before content generation proceeds. AdPilot automates decision support, but final execution remains under human control.

9. Risks & Questions Throughout Development

- Inventory Data Quality

  What happens if a seller uploads a poorly structured or incomplete CSV with missing fields, generic product names, or inconsistent formatting? The pre-GLM validation layer catches structural issues and prompts the seller to correct them before any API call is made. However, low-quality product descriptions that pass validation such as single-word product names with no category context, will limit the specificity and accuracy of GLM's recommendations. This is mitigated by instructing GLM to flag low-confidence classifications rather than fabricate specificity not present in the input, but the root issue depends on seller behavior.

- Similarity of Recommendations Across Competing Sellers

  If two sellers selling the same product category both upload similar inventories, GLM may generate near-identical strategy recommendations and ad content for both. In a competitive marketplace, this reduces the value of the recommendation for everyone. How do we ensure outputs are differentiated enough to remain competitively useful per seller?

- System Scalability During Peak Periods

  During high-traffic windows such as the week before 11.11 or Hari Raya, a large number of sellers will use AdPilot simultaneously. Z.AI's GLM API has rate limits, and a sudden spike in concurrent sessions risks delayed responses and sharp increases in API token costs at exactly the moment AdPilot is most in demand. Response caching partially addresses this by serving repeated inventory analyses from cache rather than triggering new API calls. An async queue system will also be implemented, so sessions are processed in order rather than simultaneously, preventing rate limit breaches.

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